

# AI Profile Generator

## Prompt Engineering Research Report

Project: CheckinMe

Model: Google Gemini 2.5 Flash Image (configurable)

Scope: Prompt design, comparative benchmarking & tooling

Date: 05 June 2026 (rev. 2)

Status: Directive + Prohibitions variant adopted into production

**This report documents a structured prompt engineering initiative to improve face identity preservation, skin tone fidelity, image sharpness, and background uniformity in AI-generated professional headshots. Four prompt versions are designed, benchmarked, and compared against a seven-metric quantitative framework. As of this revision (rev. 2), the Directive + Prohibitions variant (formerly experimental v3) has been adopted as the live production prompt.**

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## 1. Executive Summary

CheckinMe's AI Profile Generator uses Google Gemini to transform a user-supplied reference photograph into a set of professional headshots with varied attire. The primary quality requirement — and the most common failure mode — is **face identity preservation**: the generated image must be recognisably the same person as in the input photo.

This report documents a structured prompt engineering effort to improve generation quality across four measurable dimensions: face structural similarity, skin tone fidelity, image sharpness, and background uniformity. Four prompt versions were designed (one production baseline and three experimental variants), and a Python/Streamlit benchmarking application was built to evaluate them against a seven-metric quantitative framework augmented by human qualitative assessment.

### Key Contributions

- Identification of three structural weaknesses in the production prompt.
- Design of three experimental prompt variants, each addressing a different hypothesis about model instruction following.
- A reproducible benchmarking tool supporting side-by-side visual comparison, automated metric computation, and human rating capture.
- A configurable model selector enabling cross-model comparisons without code changes.
- A live prompt editor allowing real-time prompt modifications and A/B testing directly in the Streamlit UI.

**Outcome (rev. 2): The Directive + Prohibitions variant (formerly experimental v3) has been adopted as the production prompt. It pairs strong positive identity anchoring with an explicit negative-constraint block and a modular, sectioned layout. See section 6.7 for the prompt-engineering classification and a pros/cons assessment.**

## 2. Project Context

### 2.1 System Overview

The production generation pipeline is implemented in `AiProfileGenerateService.php` (Laravel). When a user requests headshots, the service:

- Accepts a reference image path, gender, quantity (limit), and rotation offset.
- Selects 'limit' outfit descriptions from a pool of 16 per gender, applying the offset as a circular rotation.
- Constructs one text prompt per outfit.
- Calls the Gemini multimodal endpoint (`gemini-2.5-flash-image:generateContent`) with the prompt and the base64-encoded reference image.
- Extracts the `inlineData.data` field from the API response and returns it as a base64 PNG.

In non-production environments the service short-circuits and returns random existing media URLs to avoid API cost during development.

### 2.2 Outfit Catalogue

| Gender | Pool Size | Style Range                                                           |
|--------|-----------|-----------------------------------------------------------------------|
| Male   | 16        | Black/navy/charcoal/grey suits with varied shirt and tie combinations |
| Female | 16        | Black/navy/grey/dark blazers over white and light-coloured blouses    |

All outfits are conservative corporate styles deliberately selected to minimise the risk of anatomical distortion at the neckline, a known failure mode of AI portrait generators.

### 3. Problem Statement

#### 3.1 Primary Failure Mode: Identity Drift

When a generative vision model is instructed to re-dress a subject, it faces a fundamental tension: it must synthesise new clothing while holding the face constant. Without sufficiently strong identity anchoring instructions, models tend to:

- Smooth or 'beautify' skin texture, altering the perceived person.
- Shift skin tone slightly warm or cool depending on the chosen outfit's colour temperature.
- Subtly reshape the nose, jaw, or eye area to fit a 'professional' archetype.
- Alter hair style or colour.

The combined effect is a headshot that looks professional but is not recognisably the same individual — a product failure.

#### 3.2 Secondary Failure Modes

| Failure Mode          | Visible Symptom                                          | Affected Metric                       |
|-----------------------|----------------------------------------------------------|---------------------------------------|
| Skin tone drift       | Complexion appears lighter/darker or different undertone | Skin Tone Match (20%)                 |
| Over-smoothing        | Skin looks plastic or AI-generated                       | Face SSIM (25%), Face Sharpness (20%) |
| Framing errors        | Head cropped, or too much torso shown                    | Framing Quality (10%)                 |
| Background noise      | Gradient, texture, or vignette at corners                | Background Uniformity (15%)           |
| Anatomical distortion | Floating collar, misaligned neck                         | Qualitative — Clothing Quality        |

#### 3.3 Structural Weaknesses in the Legacy Production Prompt

**Resolved in current production (rev. 2). The three issues below were identified in the legacy production prompt (Appendix E). All three are addressed by the current Directive + Prohibitions production prompt (Appendix A): identity is anchored immediately after the outfit clause, skin colour and undertone are named explicitly and reinforced by a prohibition, and the background block requires consistency from centre to all four edges of the frame. This section is retained to document the original rationale.**

Analysis of the legacy `build_production_prompt()` revealed three structural issues:

**Issue 1 — Late identity anchoring**

The production prompt opens with outfit instruction (*Generate a portrait of a professional [gender] [outfit]*) before stating face preservation requirements. This ordering implicitly signals that the primary task is outfit generation, with identity preservation as a constraint. Multimodal LLMs tend to weight earlier instructions

more heavily during synthesis.

### ***Issue 2 — Absent skin tone instruction***

The phrase 'Do not alter facial structure, skin texture, or expression' does not mention skin colour or undertone explicitly. Skin tone shift is measured at 20% of the quantitative score, making this a high-impact omission.

### ***Issue 3 — Ambiguous background specification***

'A soft, solid, professional ID card style background' is semantically underspecified. The benchmark measures background uniformity by sampling pixel variance in the four image corners. Any gradient or subtle texture that extends to the corners penalises the score; the production prompt does not instruct the model to maintain consistency to the image edges.

## 4. Technical Architecture

### 4.1 Production Service (AiProfileGenerateService.php)

User request

```
■  
■■■ getPrompts(gender, limit, offset, aspectRatio)  
■    ■■■ Selects outfits from $outfits[gender]  
■    ■■■ Applies circular offset rotation  
■    ■■■ Assembles: $commonIntro + $style + $commonFeatures  
■                      + $framingFix + $neckFix + $imageQuality  
■  
■■■ foreach $prompts as $prompt  
    ■■■ POST /v1beta/models/gemini-2.5-flash-image:generateContent  
        ■■■ parts[0]: text prompt  
        ■■■ parts[1]: inline_data (base64 PNG/JPEG)  
        → returns base64 image → stored / returned to controller
```

### 4.2 Benchmarking Application (benchmark\_app.py)

Streamlit UI

```
■  
■■■ Sidebar  
■    ■■■ Gemini API Key  
■    ■■■ Model selector (dynamic URL construction)  
■    ■■■ Gender / Outfit builder / Background selector  
■    ■■■ Experimental prompt version selector (v1 / v2 / v3)  
■  
■■■ Prompt Editor (main area)  
■    ■■■ Production text area – editable, reset button, live char count  
■    ■■■ Experimental text area – editable, reset button, live char count  
■    Auto-reloads when version/outfit/background changes (fingerprint)  
■  
■■■ Generate & Compare (button)  
■    ■■■ ThreadPoolExecutor(max_workers=2)  
■    ■■■ call_gemini_api(prod_prompt, gemini_url)  
■    ■■■ call_gemini_api(exp_prompt, gemini_url)  
■  
■■■ Side-by-Side Image Comparison  
■
```

|                                                       |                             |
|-------------------------------------------------------|-----------------------------|
| ■■■ Quantitative Metrics ■                            | bm.compute_all() → 8 scores |
| ■                                                     |                             |
| ■■■ Qualitative Rating Panel (5 criteria × 1-5 stars) |                             |
| ■■■ Final = 60% Quantitative + 40% Qualitative        |                             |

4.3 Model URL Architecture

The Gemini model endpoint follows the pattern:

```
https://generativelanguage.googleapis.com/v1beta/models/{MODEL_ID}:generateContent
```

The model ID was previously hardcoded as the module-level constant GEMINI\_URL. It is now constructed dynamically from a sidebar selectbox, enabling cross-model comparison (e.g., gemini-2.5-flash-image vs gemini-2.0-flash-exp-image-generation) without modifying source code. A custom model ID text input is also available for unreleased or preview models.



## 5. Evaluation Methodology

### 5.1 Quantitative Metrics

All metrics produce scores on a 0–100 scale, combined into an **Overall Quantitative Score** via a fixed weighted average.

| Metric                | Weight | Library      | Primary Failure Detected                                  |
|-----------------------|--------|--------------|-----------------------------------------------------------|
| Face SSIM             | 25%    | scikit-image | Identity drift — structural (SSIM on 96×96 face crop)     |
| Skin Tone Match       | 20%    | numpy        | Skin colour / undertone shift (R+G histogram correlation) |
| Face Sharpness        | 20%    | opencv       | Blur, over-smoothing (Laplacian variance, threshold 1500) |
| Background Uniformity | 15%    | numpy        | Gradients, textures, vignettes (corner std dev)           |
| Framing Quality       | 10%    | opencv       | Zoom too close or too far (face/image height ratio)       |
| Face Centering        | 5%     | opencv       | Off-centre composition (horizontal centroid offset)       |
| Exposure Quality      | 5%     | numpy        | Over/underexposure (mean luminance vs 90–175 range)       |

**Overall Quantitative Score = SSIM×0.25 + Skin×0.20 + Sharp×0.20 + BG×0.15 + Framing×0.10 + Centering×0.05 + Exposure×0.05**

### 5.2 Face Detection

Pre-processing for SSIM, Skin Tone, Sharpness, Framing, and Centering depends on locating the face. The implementation uses OpenCV's Haar cascade (haarcascade\_frontalface\_default.xml) with scaleFactor=1.1, minNeighbors=5, minSize=(40,40). If detection fails, affected metrics return a neutral score of 50 (or 0 for Framing/Centering).

### 5.3 Qualitative Assessment

| Criterion          | Question                                                           |
|--------------------|--------------------------------------------------------------------|
| Face Resemblance   | Does the face look exactly like the person in the reference photo? |
| Professional Look  | Does the overall image look like a professional headshot?          |
| Clothing Quality   | Does the clothing look natural, well-fitted, and realistic?        |
| Background Quality | Is the background clean, solid, and professional?                  |
| Overall Preference | Overall, how satisfied are you with this image?                    |

$$\text{Qualitative Score} = (\text{mean star rating} / 5) \times 100$$

5.4 Final Combined Score

$Final = 0.60 \times Quantitative\_Overall + 0.40 \times Qualitative\_Average$

| Condition           | Verdict           |
|---------------------|-------------------|
| $ Exp - Prod  < 2$  | Tie               |
| $Exp - Prod \geq 2$ | Experimental wins |
| $Prod - Exp \geq 2$ | Production wins   |

## 6. Prompt Engineering Analysis

### 6.1 Design Philosophy

Each experimental version tests a distinct hypothesis about how Gemini processes multimodal instructions:

| Version                       | Hypothesis                                                                                                          | Status                |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------|-----------------------|
| Legacy Production             | Baseline: implicit identity constraint appended after outfit instruction.                                           | Retired               |
| v1 — Photographer Role        | Assigning a domain expert persona improves instruction compliance for professional outputs.                         | Experimental          |
| v2 — Step-by-Step Identity    | Sequential processing steps with explicit metric-aligned sub-instructions improve targeted output quality.          | Experimental          |
| v3 — Directive + Prohibitions | Explicit negative constraints ( <b>X DO NOT</b> ) are more reliably honoured than equivalent positive instructions. | Adopted to production |

### 6.2 Legacy Production Prompt (Retired Baseline)

*This was the production prompt at rev. 1. It has since been replaced by the Directive + Prohibitions structure (section 6.5 and 6.7). Full text in Appendix E.*

| Attribute             | Value                                           |
|-----------------------|-------------------------------------------------|
| Approximate length    | ~430 characters / ~75 words                     |
| Instruction order     | Outfit → Identity → Framing → Anatomy → Quality |
| Skin tone mentioned   | No                                              |
| Hair preservation     | No                                              |
| Background edge spec  | No                                              |
| Negative prohibitions | No                                              |

**Strengths:** Concise, low token overhead, includes neck/anatomy fix and framing instruction.

**Weaknesses:** Outfit is the first clause — identity anchoring is secondary. No explicit skin tone, hair, age, or ethnic feature preservation. Background described as 'ID card style' — semantically ambiguous for corner uniformity. All instructions are positive; no negation list.

### 6.3 Experimental v1 — Photographer Role

| Attribute          | Value                                        |
|--------------------|----------------------------------------------|
| Approximate length | ~900 characters / ~160 words                 |
| Role assignment    | You are a professional portrait photographer |

| Attribute              | Value                              |
|------------------------|------------------------------------|
| Identity before outfit | Yes                                |
| Skin tone named        | Partially (within identity bullet) |
| Hair preservation      | Yes                                |
| Negative prohibitions  | No                                 |

**Key structural change:** Assigns a domain expert role before any task specification. Role-prompting in LLM literature shifts model behaviour toward domain-appropriate defaults. Identity block now precedes outfit instruction.

**Remaining gaps:** Skin tone and undertone not called out by name. Background uniformity to image edges not explicitly required. No negative prohibition block.

## 6.4 Experimental v2 — Step-by-Step Identity

| Attribute             | Value                                        |
|-----------------------|----------------------------------------------|
| Approximate length    | ~1,150 characters / ~200 words               |
| Structure             | 6 numbered processing steps                  |
| Skin tone named       | Yes — Step 1 first bullet, incl. undertone   |
| Background edge spec  | Yes — 'clean all the way to the image edges' |
| Sharpness instruction | Yes — 'face razor-sharp in focus'            |
| Negative prohibitions | No                                           |

**Key structural change:** Replaces free-form instruction blocks with numbered processing steps, explicitly sequencing the model to anchor identity first before applying styling.

| Metric                | Improvement Mechanism                                                                                           |
|-----------------------|-----------------------------------------------------------------------------------------------------------------|
| Skin Tone Match       | First bullet in Step 1 names skin tone and undertone explicitly; model processes identity before outfit colour. |
| Face SSIM             | Identity anchoring in Step 1 before styling reduces structural drift.                                           |
| Background Uniformity | Step 5 requires clean background 'all the way to the image edges'.                                              |
| Face Sharpness        | Step 6 instructs 'face razor-sharp in focus'.                                                                   |
| Framing               | Step 3 adds '5–10% headroom above the head' for reliable crown inclusion.                                       |

## 6.5 Experimental v3 — Directive + Prohibitions

| Attribute              | Value                                            |
|------------------------|--------------------------------------------------|
| Approximate length     | ~1,400 characters / ~240 words                   |
| Structure              | 5 headed sections + prohibitions block           |
| Lighting specification | 3-point studio (key + fill + rim)                |
| Skin tone named        | Yes — first identity bullet                      |
| Negative prohibitions  | Yes — 6 explicit <b>X DO NOT</b> rules           |
| Background edge spec   | Yes — 'consistent from center to all four edges' |

**Key structural changes:** Abandons prose/bullet style for a structured brief with section dividers and a terminal STRICT PROHIBITIONS block. Tests whether explicit negative constraints (**X DO NOT**) are more reliably honoured than equivalent positive instructions. Adds a full 3-point studio lighting specification unique to this version.

| Metric                | Improvement Mechanism                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------|
| Skin Tone Match       | First identity bullet; skin tone both positively asserted and negatively prohibited.        |
| Face SSIM             | 'Zero tolerance' framing + dual positive/negative constraints on facial geometry.           |
| Exposure Quality      | Explicit 3-point lighting with key/fill/rim spec; prohibition on HDR and cinematic grading. |
| Background Uniformity | 'Consistent from center to all four edges' targets corner-sampling metric.                  |

## 6.6 Prompt Comparison Summary

| Dimension                     | Production | v1       | v2       | v3   |
|-------------------------------|------------|----------|----------|------|
| Identity before outfit        | <b>X</b>   | ✓        | ✓        | ✓    |
| Skin tone named explicitly    | <b>X</b>   | <b>X</b> | ✓        | ✓    |
| Hair preservation             | <b>X</b>   | ✓        | ✓        | ✓    |
| Age preservation              | <b>X</b>   | <b>X</b> | ✓        | ✓    |
| Background edge uniformity    | <b>X</b>   | <b>X</b> | ✓        | ✓    |
| Sharpness / focus instruction | <b>X</b>   | ✓        | ✓        | ✓    |
| 3-point lighting spec         | <b>X</b>   | <b>X</b> | <b>X</b> | ✓    |
| Negative prohibition block    | <b>X</b>   | <b>X</b> | <b>X</b> | ✓    |
| Approx. word count            | ~75        | ~160     | ~200     | ~240 |

## 6.7 Current Production Prompt — Engineering Type, Pros & Cons

The prompt now shipping in `AiProfileGenerateService::getPrompts()` is the Directive + Prohibitions structure. It is not a single technique but a deliberate stack of several. Its dominant classification is **zero-shot, structured/modular instructional prompting with heavy constraint (negative) prompting**, applied to a text-to-image multimodal model.

### *Techniques in the stack*

| Technique                          | Where it appears in the current prompt                                                                                                                  |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Modular / compositional (template) | Final string assembled from six reusable blocks (intro, outfit, identity+background, framing, attire, lighting/quality). Only the outfit clause varies. |
| Zero-shot                          | Pure instructions — no example input/output pairs are supplied to the model.                                                                            |
| Instructional / directive          | Command voice with an explicit INPUT / OUTPUT contract ('PROFESSIONAL HEADSHOT DIRECTIVE').                                                             |
| Constraint / negative (defining)   | A terminal STRICT PROHIBITIONS block of X Do NOT rules, plus the IDENTITY — ZERO TOLERANCE framing.                                                     |
| Delimiter / sectioned formatting   | ■■■■ SECTION ■■■■ headers segment the prompt into semantic regions.                                                                                     |
| Emphasis / repetition (priming)    | Identity preservation is stated up front and restated as a prohibition; CAPS and → arrows weight critical tokens.                                       |
| Parameterized variation            | limit / offset rotate a 16-item outfit list for controlled diversity from one template.                                                                 |

**One-line label for citation:** *structured zero-shot instructional prompting with negative constraints (modular, template-based).*

### *Pros*

- **Consistency & reproducibility** — fixed blocks make every headshot read as if from the same studio; only the outfit changes.
- **Maintainability (DRY)** — six isolated components let one lever be tuned without touching the others.
- **Controlled diversity** — offset/limit rotation produces unique-but-on-brand variants deterministically.
- **Strong guardrails** — explicit Do NOT prohibitions target known failure modes (face drift, skin smoothing, added accessories, HDR stylization).
- **Self-documenting** — section delimiters make intent legible to the model and to future maintainers.
- **Low cost / low complexity** — zero-shot needs no example corpus, fine-tuning, or retrieval pipeline.

### *Cons*

- **Verbosity → token cost & dilution** — long, repetitive prompts can cause image models to weight tokens unevenly and average out key instructions.
- **Negative-prompt unreliability** — image models often honour 'Do NOT X' poorly; naming a concept can raise its chance of appearing. Native negative-prompt fields are usually more reliable.
- **No reference grounding for style** — identity rides on the single input image plus words; few-shot or stronger conditioning typically locks identity better than 'ZERO TOLERANCE' phrasing.
- **Model-coupling / brittleness** — wording is tuned to gemini-2.5-flash-image; a model swap can silently degrade results.

- **Soft / unenforced parameters** — aspectRatio is passed as prompt text only, not via API config, so the model may ignore it.
- **Limited semantic diversity** — variation is only the outfit string; pose, expression, and crop are frozen, so outputs can feel templated.
- **Emphasis overuse** — heavy CAPS and symbols have diminishing returns and can hurt if the model over-anchors on 'shouted' tokens.

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## 7. Benchmarking Tool Design

### 7.1 Live Prompt Editor

The original application displayed prompts in a read-only expander. This was replaced with a live prompt editor — two side-by-side editable text areas with the following behaviour:

- **Auto-reload on fingerprint change:** A fingerprint string encodes gender | custom\_outfit | experimental\_outfit | background\_style | exp\_version\_key. When any sidebar selection changes, both text areas reload with the updated preset.
- **Manual override:** Within a constant fingerprint, user edits persist across Streamlit re-runs. Exactly the edited text is sent to the API.
- **Reset buttons:** ■ Reset restores the text area to the current preset without changing other state.
- **Live statistics:** Character and word count displayed beneath each editor.

### 7.2 Experimental Version Selector

A radio widget in the sidebar presents the three experimental versions. Selecting a version changes the fingerprint, triggers auto-reload of the experimental text area, and labels the result image header with the version name for unambiguous identification.

### 7.3 Configurable Model Selector

The Gemini model is now constructed dynamically from a sidebar selectbox. `call_gemini_api` accepts `url` as an explicit parameter (defaulting to `GEMINI_URL` for backwards compatibility). Both production and experimental calls use the same model, ensuring fair comparison.

### 7.4 Parallel Generation

Both prompts are submitted concurrently using `ThreadPoolExecutor(max_workers=2)`, minimising total wall-clock time and ensuring that any temporal variation in the Gemini API (temperature, model load) affects both prompts equally.



## 8. Results

This section will be populated after benchmark runs are completed. The framework is ready for data collection.

### 8.1 Data Collection Protocol

For statistically meaningful results, each prompt version should be tested with:

- Minimum 3–5 different reference photos (diverse skin tones, genders, ages).
- At least 3 generations per photo per version, given Gemini's stochastic nature.
- Qualitative ratings completed immediately after each generation while the reference image is visible.

### 8.2 Expected Results (Hypothesis)

| Metric                | Expected Ranking          |
|-----------------------|---------------------------|
| Face SSIM             | v3 ≥ v2 > v1 > Production |
| Skin Tone Match       | v3 ≥ v2 > v1 > Production |
| Face Sharpness        | v2 ≈ v3 > v1 > Production |
| Background Uniformity | v2 ≈ v3 > v1 > Production |
| Framing Quality       | v2 ≥ v1 ≈ v3 ≈ Production |
| Exposure Quality      | v3 > v2 ≈ v1 ≈ Production |
| Overall Score         | v3 ≥ v2 > v1 > Production |

**Note:** Longer prompts are not always better. There is a risk that v3's structured format (Unicode dividers, ✕ symbols) is less well-handled by the model than plain prose, which could cause unexpected regressions. This hypothesis must be validated empirically.

### 8.3 Results Table (to be filled)

| Run | Reference Photo | Gender | Version | Model | SSIM | Skin | Sharp | BG | Overall |
|-----|-----------------|--------|---------|-------|------|------|-------|----|---------|
| —   | —               | —      | —       | —     | —    | —    | —     | —  | —       |

---

## 9. Recommendations

### 9.1 Immediate

- Run the benchmark with 3+ reference photos before making any prompt changes to production. Never change a production prompt based on visual intuition alone.
- Use gemini-2.5-flash-image for all runs in the initial benchmark pass to ensure model is not a confounding variable.
- Rate qualitative scores immediately after generation — recency bias affects scoring accuracy if ratings are done in batch later.

### 9.2 Short-Term (pending benchmark results)

- If v2 or v3 outperforms production on Face SSIM + Skin Tone Match without regressing on Framing/Exposure, adopt the winner as the new production prompt in AiProfileGenerateService.php.
- Consider adding skin tone mention to the production prompt as a minimal low-risk improvement even before full experiment results are available, since it addresses a clear gap with no downside.

### 9.3 Medium-Term

- Replace Haar cascade with a deep-learning face detector (MTCNN or RetinaFace). Haar cascade fails on non-frontal faces and profiles.
- Add FaceNet / ArcFace cosine similarity as an 8th metric. SSIM measures structural patterns but is not identity-aware — embedding-based identity distance is a more principled measure.
- Average metrics across 3 generations per prompt to reduce per-run stochasticity. A single bad generation can skew Overall Score by 5–10 points.
- Add CLIP scoring ('professional business headshot' anchor text) as a semantic quality metric independent of the reference photo.

### 9.4 Long-Term

- Consider prompt personalisation by skin tone range: darker skin tones are more susceptible to brightening drift by the model.
- Evaluate Gemini 2.0 Flash Experimental as a model upgrade candidate using the configurable model selector added in this work.

# 10. Appendix — Full Prompt Texts

## A Production Prompt (Current — Directive + Prohibitions)

|                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------|
| PROFESSIONAL HEADSHOT DIRECTIVE                                                                                                          |
| INPUT: Reference photograph of a real person                                                                                             |
| OUTPUT: Professional {gender} corporate headshot, {outfit}.                                                                              |
| IDENTITY — ZERO TOLERANCE FOR CHANGES                                                                                                    |
| The person in the generated image must be the EXACT SAME PERSON as in the reference photo. Treat the face as a locked, uneditable asset: |
| Skin color and undertone: match exactly — same warmth, depth, and tone                                                                   |
| Skin texture: preserve all natural texture, pores, fine lines — no AI smoothing                                                          |
| Facial geometry: same bone structure, proportions, and all features                                                                      |
| Eyes: same color, shape, spacing, and brow arch                                                                                          |
| Hair: same color and style as shown                                                                                                      |
| Age: no de-aging or aging — keep the subject's natural age                                                                               |
| BACKGROUND                                                                                                                               |
| Soft neutral grey — clean, uniform, professional studio backdrop.                                                                        |
| Completely free of texture, objects, or environmental context.                                                                           |
| Background must be consistent from center to all four edges of the frame.                                                                |
| FRAMING & POSE                                                                                                                           |
| Head-to-mid-chest crop — full crown of head with small headroom margin                                                                   |
| Face centered horizontally                                                                                                               |
| Upright, professional posture — relaxed shoulders, no unnatural tilt                                                                     |
| ATTIRE                                                                                                                                   |
| Garment must be sharp, properly fitted, and wrinkle-free.                                                                                |
| Collar and neckline sit naturally against the neck — no gaps or floating fabric.                                                         |
| Neck and shoulder anatomy anatomically correct and proportional.                                                                         |
| LIGHTING                                                                                                                                 |
| 3-point studio setup:                                                                                                                    |
| Key light — upper-left at 45°, soft box diffused                                                                                         |
| Fill light — right side, softer intensity to open shadows                                                                                |
| Rim / hair light — subtle, from behind, to separate subject from background                                                              |

Result: even, shadow-minimized illumination; true-to-life color; no blown highlights on skin.

TECHNICAL OUTPUT

- Photorealistic DSLR photograph – high resolution, no compression artifacts
- 85mm portrait lens equivalent at f/2.2 – face tack-sharp, background subtly soft
- No artistic filters, no painterly or HDR effects, no stylization

STRICT PROHIBITIONS

- Do NOT generate a different face – must be recognizable as the reference person
- Do NOT smooth, beautify, or retouch skin
- Do NOT change skin tone, hair color, or eye color
- Do NOT alter facial proportions or make the person look any different
- Do NOT add accessories, props, or background elements not specified
- Do NOT apply cinematic, HDR, or stylized color grading

B Experimental v1 — Photographer Role

You are a professional portrait photographer. Using the uploaded reference photo, generate a photorealistic headshot portrait with the following specifications:

IDENTITY PRESERVATION (CRITICAL):

- The subject's face must be IDENTICAL to the person in the uploaded image – same skin tone, eye color, nose shape, jawline, and every facial feature
- Do NOT alter, enhance, smooth, or modify the face in any way
- Preserve exact skin texture, any facial marks, and natural expression
- Do not change hair color, hair style, or eyebrow shape

SUBJECT & COMPOSITION:

- Professional {gender} portrait
- {outfit}
- Framing: from the crown of the head to the mid-chest, face centered horizontally
- Natural, confident, upright posture – shoulders squared to camera

CLOTHING & ANATOMY:

- Clothing must be sharp, properly fitted, wrinkle-free
- Collar and neckline must sit naturally – no gaps, floating fabric, or distortion
- Neck and shoulders must be anatomically correct and proportional

BACKGROUND:

- {background}, clean professional background (LinkedIn / ID card portrait style)
- No props, no distractions – soft gradient or flat solid color only

#### PHOTOGRAPHY STYLE:

- Ultra-photorealistic – must look like a real photograph taken in a professional studio
- No artistic filters, no painterly effects, no 3D rendering
- Soft box studio lighting: main light slightly above-left, subtle fill from the right
- Face in sharp focus, background softly diffused (shallow depth of field)
- Color grading: neutral, true-to-life – no heavy warm/cool toning

OUTPUT: A single portrait image matching all specifications above.

## C Experimental v2 — Step-by-Step Identity

You are processing a real reference photograph to generate a professional corporate headshot. The face in the reference photo must appear EXACTLY in the output.

#### STEP 1 – ANCHOR THE IDENTITY (highest priority):

Study every detail of the face in the uploaded photo before generating:

- Skin tone and undertone (warm / cool / neutral) – match precisely
- Skin texture: pores, fine lines, any marks or freckles – do not smooth away
- Eye color, shape, and spacing
- Nose shape, lip shape, jawline, and cheekbone structure
- Hair: exact color, texture, and current style as shown
- Apparent age – do not de-age or age the person
- Ethnicity and all distinguishing features

This face must appear UNCHANGED in the output.

#### STEP 2 – APPLY PROFESSIONAL STYLING:

Subject: {gender} professional     Attire: {outfit}

#### STEP 3 – COMPOSITION:

- Framing: crown of head to mid-chest, 5-10% headroom above the head
- Face horizontally centered in frame
- Upright posture, shoulders relaxed, confident neutral expression

#### STEP 4 – CLOTHING & ANATOMY:

- Clothing crisp, well-tailored, wrinkle-free
- Collar sits flat against neck – no floating fabric, no gaps
- Neck and shoulder anatomy natural and proportional

STEP 5 – BACKGROUND:

{background} – completely uniform solid or very subtle gradient  
Must be clean all the way to the image edges – no texture or environmental elements

STEP 6 – LIGHTING & FOCUS:

Key light at 45° upper-left, gentle fill from right  
Even illumination – no harsh chin or nose shadows  
Face razor-sharp in focus; background may have slight depth-of-field softness  
Ultra-photorealistic – indistinguishable from a real studio photograph

OUTPUT: Single portrait image exactly matching the above specifications.

D Experimental v3 — Directive + Prohibitions

PROFESSIONAL HEADSHOT DIRECTIVE

INPUT: Reference photograph of a real person  
OUTPUT: Professional {gender} corporate headshot

IDENTITY – ZERO TOLERANCE FOR CHANGES

The person in the generated image must be the EXACT SAME PERSON as in the reference photo. Treat the face as a locked, uneditable asset:

Skin color and undertone: match exactly – same warmth, depth, and tone  
Skin texture: preserve all natural texture, pores, fine lines – no AI smoothing  
Facial geometry: same bone structure, proportions, and all features  
Eyes: same color, shape, spacing, and brow arch  
Hair: same color and style as shown  
Age: no de-aging or aging – keep the subject's natural age

ATTIRE

{outfit} | Garment must be sharp, properly fitted, and wrinkle-free.  
Collar and neckline sit naturally – no gaps or floating fabric.

FRAMING & POSE

Head-to-mid-chest crop – full crown of head with small headroom margin  
Face centered horizontally  
Upright, professional posture – relaxed shoulders, no unnatural tilt

BACKGROUND

{background} – clean, uniform, professional studio backdrop

Completely free of texture, objects, or environmental context.

Background must be consistent from center to all four edges of the frame.

#### LIGHTING (3-point studio setup)

Key light – upper-left at 45°, soft box diffused

Fill light – right side, softer intensity to open shadows

Rim / hair light – subtle, from behind, to separate subject from background

#### TECHNICAL OUTPUT

Photorealistic DSLR photograph – high resolution, no compression artifacts

85mm portrait lens equivalent at f/2.2 – face tack-sharp, background soft

No artistic filters, no painterly or HDR effects, no stylization

#### STRICT PROHIBITIONS

Do NOT generate a different face

Do NOT smooth, beautify, or retouch skin

Do NOT change skin tone, hair color, or eye color

Do NOT alter facial proportions

Do NOT add accessories, props, or background elements not specified

Do NOT apply cinematic, HDR, or stylized color grading

## E Legacy Production Prompt (pre-2026-06)

Generate a portrait of a professional {gender} {outfit}.

STRICTLY PRESERVE the subject's original face. The output must look exactly

like the person in the uploaded image. Do not alter facial structure, skin

texture, or expression. The body should be centered with a natural, professional

pose. Background must be a soft, solid, professional ID card style background.

Frame the image from the top of the head to the mid-chest. Ensure the neck and

shoulders are anatomically correct and proportional. The clothing must fit

naturally around the neck/collar area without floating or weird cuts.

PHOTOREALISTIC style. The image must look like a high-end photograph.

No cartoonish, 3D render, or filtered looks. Lighting should be natural studio

lighting.